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Silverio Martínez Andrés

Algebra Lineal

Grupo 17

Semestre 2022-2

# Tarea 4

Sean el espacio vectorial $\mathbb{R}^3$ con producto interno usual

$$W = \{ (x, y, z) \mid 3y - 2z - x = 0;\ x, y, z \in \mathbb{R} \}$$

un subespacio de $\mathbb{R}^3$:

a) Determinar el complemento ortogonal.

b) Expresar el vector $(9, 1, 4)$ como $\overline{x} = \overline{w} + \overline{w}^{\perp}$

Solución de a)

$3y - 2z - x$

$-x = -3y + 2z \implies x = 3y - 2z$

$$W = \{ (3y - 2z,\ y,\ z) \mid y, z \in \mathbb{R} \}$$

$$\left.\begin{matrix} y = 1 \\ z = 0 \end{matrix}\right\} \implies (3, 1, 0)$$

$$\left.\begin{matrix} y = 0 \\ z = 1 \end{matrix}\right\} \implies (-2, 0, 1)$$

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$B = \{(3,1,0)\ (-2,0,1)\}$

$W^{\perp} = \{\bar{v} \in V \mid (\bar{v} \mid \bar{w}) = \emptyset ;\ \bar{w} \in W\}$

$W^{\perp} = \{(x,y,z) \in \mathbb{R}^3 \mid (x,y,z) \mid \begin{matrix}(3,1,0) \\ (-2,0,1)\end{matrix} = \emptyset$

$(\bar{x} \mid \bar{y}) = x_1 y_1 + x_2 y_2 + x_3 y_3$

$(\underset{x_1}{x}, \underset{x_2}{y}, \underset{x_3}{z}) \mid (\underset{y_1}{3}, \underset{y_2}{1}, \underset{y_3}{0}) = 3x + y = \emptyset$

$\underline{y = -3x}_{//}$

$(x, y, z) \mid (-2, 0, 1) = -2x + z = \emptyset$

$\underline{z = 2x}_{//}$

$W^{\perp} = \{(x, y, z) \in \mathbb{R}^3\}$

$x = x$

$y = -3x$

$z = 2x$

$\therefore W^{\perp} = \{(x, -3x, 2x) \mid x \in \mathbb{R}\}$ $\Rightarrow a)_{//}$

Complemento ortogonal

Escaneado con CamScanner

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Comprobación

$(\bar{v} \mid \bar{w}) = \emptyset$

$(\underset{x_1}{x}, \underset{x_2}{-3x}, \underset{x_3}{2x}) \mid (\underset{y_1}{3y-2z}, \underset{y_2}{y}, \underset{y_3}{z})$

$(\bar{x} \mid \bar{y}) = x_1 y_1 + x_2 y_2 + x_3 y_3$

$(x)(3y-2z) + (-3x)(y) + (2x)(z) = \emptyset$

$\cancel{3xy} - \cancel{2xz} - \cancel{3xy} + \cancel{2xz} = \emptyset$

$\emptyset = \emptyset$

$\therefore$ Si cumple //

Solución inciso b)

$\bar{x} = \bar{w} + \bar{w}^{\perp}$

$(9, 1, 4) = (3y - 2z, y, z) + (x, -3x, 2x)$

$(9, 1, 4) = (x + 3y - 2z, -3x + y, 2x + z)$

$$\left.\begin{aligned} x + 3y - 2z &= 9 \\ -3x + y &= 1 \\ 2x + z &= 4 \end{aligned}\right\} \text{Si de ecu}$$

$$\left(\begin{array}{ccc|c} 1 & 3 & -2 & 9 \\ -3 & 1 & 0 & 1 \\ 2 & 0 & 1 & 4 \end{array}\right) \underset{(-3)R_1}{\sim} \left(\begin{array}{ccc|c} -3 & -9 & 6 & -27 \\ -3 & 1 & 0 & 1 \\ 2 & 0 & 1 & 4 \end{array}\right) \underset{\substack{R_1 - R_2 = R_2 \\ (-1/3)R_1}}{\sim}$$

$$\left(\begin{array}{ccc|c} 1 & 3 & -2 & 9 \\ 0 & 10 & -6 & 28 \\ 2 & 0 & 1 & 4 \end{array}\right) \underset{(2)R_1}{\sim} \left(\begin{array}{ccc|c} 2 & 6 & -4 & 18 \\ 0 & 10 & -6 & 28 \\ 2 & 0 & 1 & 4 \end{array}\right) \sim$$

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$R_1 - R_3 = R_3 \quad (1/2)R_1$

$$\left(\begin{array}{ccc|c} 1 & 3 & -2 & 9 \\ 0 & 10 & -6 & 28 \\ 0 & -6 & 5 & -14 \end{array}\right) \underset{(3)R_2}{\overset{(\frac{1}{10})R_2}{\sim}} \left(\begin{array}{ccc|c} 1 & 3 & -2 & 9 \\ 0 & 3 & -9/5 & 42/5 \\ 0 & -6 & 5 & -14 \end{array}\right) \underset{(-6)R_2}{\overset{R_1 - R_2 = R_1}{\sim}}$$

$$\left(\begin{array}{ccc|c} 1 & 0 & -1/5 & 3/5 \\ 0 & -6 & 18/5 & -84/5 \\ 0 & -6 & 5 & -14 \end{array}\right) \underset{(-1/6)R_2}{\overset{R_3 - R_2 = R_3}{\sim}} \left(\begin{array}{ccc|c} 1 & 0 & -1/5 & 3/5 \\ 0 & 1 & -3/5 & 14/5 \\ 0 & 0 & 7/5 & 14/5 \end{array}\right)$$

$(7/5)R_3 \quad (-1/5)R_3$

$$\left(\begin{array}{ccc|c} 1 & 0 & -1/5 & 3/5 \\ 0 & 1 & -3/5 & 14/5 \\ 0 & 0 & -1/5 & -2/5 \end{array}\right) \underset{(-5)R_3}{\overset{R_3 - R_1 = R_1}{\sim}} \left(\begin{array}{ccc|c} 1 & 0 & 0 & \\ 0 & 1 & -3 & 14/5 \\ 0 & 0 & 1 & 2 \end{array}\right) \overset{(-3/5)R_3}{\sim}$$

$R_3 - R_2 = R_2$

$(-3/5)/R_3$

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 4 \\ 0 & 0 & 1 & 2 \end{array}\right) \quad \therefore \quad \begin{array}{l} x = 1 \\ y = 4 \\ z = 2 \end{array}$$

$(9,1,4) = (3y - 2z, y, z) + (x, -3x, 2x)$

$(9,1,4) = (3(4) - 2(2), 4, 2) + (1, -3(1), 2(1))$

$\therefore \bar{x} = \bar{w} + \bar{w}^{\perp} \Rightarrow (9,1,4) = (8,4,2) + (1,-3,2)$

Comprobación

$(9,1,4) = (8+1, 4-3, 2+2)$ $\quad \therefore \checkmark$ Si cumple

$(9,1,4) = (9,1,4)$